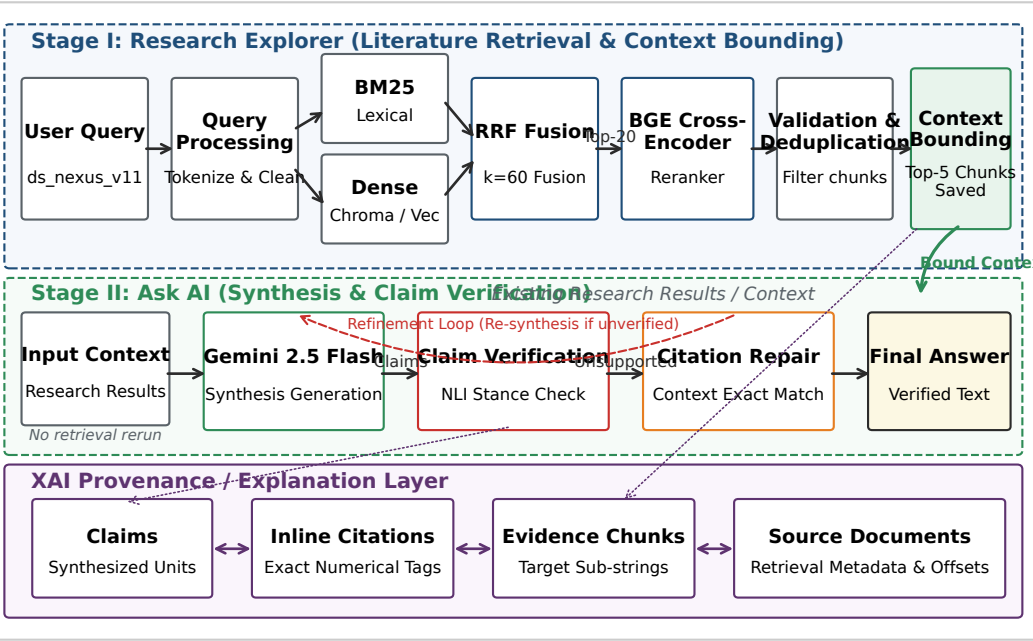
Stage I: Research Explorer (Literature Retrieval & Context Bounding)
User Query
ds_nexus_v11
Query Processing
Tokenize & Clean
BM25
Lexical
Dense
Chroma / Vec
RRF Fusion
k=60 Fusion
Top-20
BGE Cross-Encoder
Reranker
Validation & Deduplication
Filter chunks
Context Bounding
Top-5 Chunks Saved
Bound Context
Stage II: Ask AI (Synthesis & Claim Verification)
Validated Research Results / Context
Input Context
Research Results
No retrieval rerun
Gemini 2.5 Flash
Synthesis Generation
Refinement Loop (Re-synthesis if unverified)
Claim Verification
NLI Stance Check
Claims
Unsupported
Citation Repair
Context Exact Match
Final Answer
Verified Text
XAI Provenance / Explanation Layer
Claims
Synthesized Units
Inline Citations
Exact Numerical Tags
Evidence Chunks
Target Sub-strings
Source Documents
Retrieval Metadata & Offsets